

Lab 7 Report: Usability Evaluation

MMAC Study Hub and Aura Screen

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1 Introduction

We evaluated two products from the Multimodal Affective Computing (MMAC) study. The **Study Hub** is a website that introduces the study and showcases its thermal and ECG data in interactive dashboards. **Aura Screen** is a patient health-screening app for phones and screening kiosks: sign up, scan at a station, read the report.

Three problems drove this evaluation. Aura Screen was built for a desktop browser, so its phone/kiosk view could only be shown through browser developer tools, and screens needed scrolling. The hub linked to separate Streamlit sensor dashboards that stopped launching after a while because of deprecated dependencies. And the project’s goal moved from *recording* data to *showcasing* it, which the recording dashboards did not fit.

2 Test plan

Tasks.

- **T1 (Aura Screen, phone):** create an account, finish onboarding, start a scan and read your heart rate on the report.
- **T2 (Study Hub, laptop):** find the ECG data, compare heart rate at rest vs city driving, and start a data-access request.

T1 is the app’s core end-to-end journey; T2 covers the hub’s two jobs, showing data and controlling its release, and requires reading a chart.

Users. Five people from outside the team: non-technical users (closest to patients) and technical students (closest to researchers).

Procedure. About 12 minutes each: consent, a think-aloud warm-up, T1, T2, the SUS questionnaire and two debrief questions. One of us facilitates; the other observes and fills in the test form.

Metrics. Completion (effectiveness), time on task (efficiency), errors (where the design fails), think-aloud confusion points (why), and SUS (standard satisfaction score; 68 is average).

3 Findings

3.1 User testing

	T1 Aura Screen		T2 Study Hub	
	Done	Time (s)	Done	Time (s)
P1	—	—	—	—
P2	—	—	—	—
P3	—	—	—	—
P4	—	—	—	—
P5	—	—	—	—
Rate / mean	— / 5	—	— / 5	—
Errors (mean)	—		—	

Mean SUS: **[from forms]**. Confusion points: **[from forms]**.

3.2 Heuristic evaluation

Severity: 0 none, 1 cosmetic, 2 minor, 3 major, 4 catastrophe. A = Aura Screen, W = website. H1 system status, H2 real world, H4 consistency, H5 error prevention, H6 recognition, H7 flexibility, H8 minimalist design, H9 error recovery, H10 help.

Aura Screen				
A1	Phone/kiosk view only reachable through browser dev tools.	H7, H4		3
A2	Core screens scrolled on a phone (e.g. Scan Hub +454 px on 390×844).	H8		3
A3	Report: third heart-rate tile cut off at phone width.	H8		3
A4	Navigation at the top, out of thumb reach.	H4		2
A5	13 px inputs make iOS zoom on tap.	H4		2
A7	Kiosk screens left large empty areas.	H8		2
A6	Hover styles “stick” after a tap.	H1		1

Study	Hub website		
W1	Dashboard links dead when their servers were down or failed to launch (deprecated versions); no explanation shown.	H9, H1	4
W2	Dashboards were recording tools; users need to view data.	H2	3
W3	Separately styled Streamlit apps broke consistency.	H4	3
W4	Site failed to start (stray build cache); no run instructions.	H10	3
W5	Placeholder “lorem ipsum” text.	H8	2
W6	Data offered as open downloads, no terms.	H5	2
W7	Charts work with a mouse only.	H7	2
W9	Request relies on an email app being set up.	H9	2
W8	Widget dragging is weakly signposted.	H6	1
W10	Thermal sample temperatures are illustrative.	H2	1

3.3 Accessibility check

Measured on the live site by computing the WCAG contrast of every visible text style against its real background (AA: 4.5:1, or 3:1 for large text).

Check	Result	
Contrast, website	9 of 24 styles fail on home: inactive items 2.30:1, grey text on cards 4.09:1.	Fail
Contrast, charts	Axis labels 3.1:1 at 10 px (light mode).	Fail
Contrast, Aura	Body 9.7:1, buttons 10.8:1 pass; captions 2.30:1, accent 2.53:1.	Fail
Text size	Smallest 9 px; several styles under 12 px.	Part.
Keyboard	Widgets focusable and movable (Alt+arrows); dialog traps focus, Esc closes; charts not keyboard-operable.	Part.
Labels	All images have alt text; one icon button (theme toggle) has no name.	Part.
Touch targets	Aura tabs 50 px, inputs 48 px.	Pass

3.4 Live demo

Aura Screen on a laptop: the old build needs dev-tools emulation and Scan Hub scrolls; the phone/kiosk frame fits the same screen without scrolling (A1, A2).

4 Improvement plan

Ranked by severity. “Ready” = prototype built, to verify with users next lab.

Issues	Planned change	Status
W1–W3	Replace the Streamlit links with data-showcase pages built into the site (draggable widget dashboards, same analysis code).	Ready
A1, A2	Phone/kiosk-first Aura Screen: device-sized frame on desktop, full screen on devices, core screens fit without scrolling.	Ready
A3, W4	Fix the report card; one-command launcher and run instructions.	Ready
Contrast	Raise grey and caption text to $\geq 4.5:1$ in both products.	Planned
A4, A5, A7	Bottom tab bar, 16 px inputs, kiosk layouts that fill the screen.	Ready
W6	Access request with a data-use agreement instead of downloads.	Ready
W7, keyboard	Keyboard access and a data table for each chart.	Planned
Labels, size	Name the theme toggle; 12 px minimum text.	Planned

[Add any new issues from user testing and re-rank.]